

Inferred Solution of Unsteady Stokes Flow using Physics Informed Gaussian Process

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Chapter 1

Introduction

1.1 Unsteady Stokes Flow

The motion of an incompressible Newtonian fluid is governed by the Navier-Stokes equations, obtained from the Cauchy momentum equation together with the incompressibility constraint (Bird et al., 2002),

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \eta \nabla^2 \mathbf{u} + \mathbf{f}_{ext}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

where $\mathbf{u}$ is the velocity field, p the pressure, ρ the fluid density, η the dynamic viscosity, and $\mathbf{f}_{ext}$ an external body force. The relative importance of the inertial (advective) term $(\mathbf{u} \cdot \nabla) \mathbf{u}$ against the viscous term $\eta \nabla^2 \mathbf{u}$ is measured by the Reynolds number, $Re = \rho U L / \eta$, for characteristic velocity U and length scale L (Bird et al., 2002). For $Re \ll 1$, the advective term can be neglected, reducing (1) to the unsteady Stokes equations,

$$\rho \frac{\partial \mathbf{u}}{\partial t} = -\nabla p + \eta \nabla^2 \mathbf{u} + \mathbf{f}_{ext}, \quad \nabla \cdot \mathbf{u} = 0. \quad (2)$$

Note that, unlike the steady Stokes flow condition commonly used to describe low-Reynolds-number flows ($Re \approx 0$), the time-derivative term $\rho \partial \mathbf{u} / \partial t$ is retained here, while dropping the advective nonlinearity term. Equation (2)

therefore remains valid for flows driven by time-dependent boundary conditions or external force, which is the regime of interest in this work. For example, the transient start-up of pressure-driven flow, oscillatory or pulsatile forcing, and the low-Reynolds-number swimming or transport of particles in a viscous medium, where the surrounding fluid does not reach a steady state on the timescale of the motion of interest.

Physics-informed Gaussian processes (PIGP) have recently been proposed as a probabilistic alternative to mesh-based solvers (e.g. FEM) and neural-network-based solvers (e.g. PINNs) for such flow problems. Molina et al. (2023) introduced the *Stokesian process* (SP), in which the linearity of the steady Stokes equations $0 = -\nabla p + \eta \nabla^2 \mathbf{u} + \mathbf{f}_{ext}$, $\nabla \cdot \mathbf{u} = 0$, is used to construct a Gaussian process whose covariance kernel is derived directly from the governing equations, rather than imposed as a soft penalty in a loss function as is done for PINNs. This construction guarantees that the inferred velocity and pressure fields satisfy the governing equations exactly at every training point, and provides a principled estimate of prediction uncertainty, at the cost of solving a purely *steady* flow problem.

Independently, Raissi et al. (2018) introduced *numerical Gaussian processes*, in which a Gaussian process prior is placed on the solution of a time-dependent partial differential equation at each time step, with the covariance kernel obtained by encoding a chosen time-discretization scheme (e.g. backward Euler, Runge–Kutta) directly into the kernel. The solution is then advanced in time by treating the *posterior mean* at time step $t - 1$ as *artificial training data* for the inference at time step t , avoiding any spatial mesh or discretization of the spatial differential operators.

This work combines these two approaches to solve the *unsteady* Stokes flow problem of (2): the spatial structure of the Gaussian process kernel follows the Stokesian process construction of Molina et al. (2023), while the time-marching scheme, using a backward Euler discretization and artificially generated training data at each step, follows Raissi et al. (2018). The full derivation of the combined kernel and inference procedure is given in Chapter 2, and is applied to pressure-driven flow between parallel plates and through a sinusoidal channel in Chapter 3.

1.2 Gaussian Process Regression

Gaussian Process or GP are defined by a series of infinite value that have a joint Gaussian distribution. A GP for function $f(x)$, with the input locations at $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, will provide values for $\mathbf{f}(\mathbf{X}) = [\mathbf{f}(\mathbf{x}_1), \mathbf{f}(\mathbf{x}_2), \dots, \mathbf{f}(\mathbf{x}_N)]$. The joint distribution is in the form of a multi-variate Gaussian

$$p(\mathbf{f}(\mathbf{X})|\mathbf{X}, \boldsymbol{\theta}, \mu(\cdot), k(\cdot, \cdot)) = \mathcal{N}(\mu(\mathbf{X}), k(\mathbf{X}, \mathbf{X}; \boldsymbol{\theta})) \quad (3)$$

μ and k represent the average function and kernel function respectively

$$\mu(\mathbf{X}) = \langle f(\mathbf{X}) \rangle \equiv \langle f \rangle(\mathbf{x}) \quad (4)$$

$$k(\mathbf{X}, \mathbf{X}; \theta) = \langle \delta f(\mathbf{x}) \delta f(\mathbf{x}') \rangle \equiv \langle \delta f \delta f(\mathbf{x}, \mathbf{x}') \rangle \quad (5)$$

where θ is the hyperparameter of the kernel function. The kernel function used was a square exponential kernel.

$$k(x, x') = \exp \left(-\frac{(x - x')^2}{2l^2} + \gamma \right) \quad (6)$$

Here, l and γ are the hyperparameters to the kernel function, controlling the length scale and amplitude of the correlations, respectively (Molina et al., 2023). To capture the dimensionality of the problem, derived kernels were utilized such as the product kernel, to calculate the kernel function.

$$k^\times(\mathbf{x}, \mathbf{x}') = k(x, x') \times k(y, y') \quad (7)$$

We can learn the function by inferring the value at each test positions $\mathbf{X}_*$. For a function f ,

$$\begin{bmatrix} \mathbf{f}^\star \\ \mathbf{f} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \boldsymbol{\mu}^\star \\ \boldsymbol{\mu} \end{bmatrix}, \begin{bmatrix} \mathbf{K}_{\star\star} & \mathbf{K}_\star \\ \mathbf{K}_\star^T & \mathbf{K} \end{bmatrix} \right) \quad (8)$$

the distribution of $\mathbf{f}_\star \mid \mathbf{f} \sim \mathcal{N}(\boldsymbol{\mu} + \mathbf{K}_\star \mathbf{K}^{-1} \delta \mathbf{f}, \mathbf{K}_{\star\star} - \mathbf{K}_\star \mathbf{K}^{-1} \mathbf{K}_\star^T)$. Following Molina et al. (2023), the kernel hyperparameters $\boldsymbol{\theta}$ are treated via a maximum a posteriori (MAP) estimate rather than full Bayesian marginalization,

$$\boldsymbol{\theta}^\star = \arg \max_{\boldsymbol{\theta}} [\log p(\boldsymbol{\theta} \mid \mathbf{X}, f(\mathbf{X}), I)] \quad (9)$$

where the posterior is given by $p(\boldsymbol{\theta} \mid \mathbf{X}, f(\mathbf{X}), I) \propto p(f(\mathbf{X}) \mid \mathbf{X}, \boldsymbol{\theta}, I) p(\boldsymbol{\theta} \mid I)$, following the same construction as Molina et al. (2023).

At each timestep, the optimization is carried out using a Nelder–Mead downhill simplex method over 500 iterations, matching the approach of Molina et al. (2023). The nature of our hyperparameters, and stabilization factor of the optimization method will be discussed in full details in Chapter 2

Finally, this framework is extended from an arbitrary function f to the velocity and pressure fields of (2) by exploiting the closure of Gaussian processes under linear operators: since the Stokes momentum balance and continuity equations are linear in $\mathbf{u}$ and p , any linear combination of their derivatives (e.g. ∇p , $\nabla^2 \mathbf{u}$, $\nabla \cdot \mathbf{u}$) is itself a Gaussian process, jointly correlated with $\mathbf{u}$ and p through kernels obtained by applying the corresponding differential operators to $k^\times$ (Molina et al., 2023). This is what allows the governing equations to be imposed *exactly* at a finite set of training points, rather than approximately through a loss function. The full construction, including its extension to the unsteady case via backward Euler time-stepping, is given in the next chapter.

Chapter 2

Methodology

The biggest challenge in solving this unsteady flow system is capturing the time propagation of velocity. This chapter first derives a backward Euler time discretization of the unsteady Stokes equations (Section 2.1), then builds the physics-informed Gaussian process kernel that encodes both this time discretization and the steady Stokes equations, following Molina et al. (2023) (Section 2.2). Section 2.3 specializes this construction to the periodic channel-flow geometry used in this work, including how uncertainty from earlier time steps is propagated forward (Raissi et al. (2018)) and how the kernel hyperparameters are optimized. Finally, Section 2.6 describes the full time-marching inference algorithm.

2.1 Time Discretization

There are several methods to capture time-stepping in the governing equations, but for the particular case considered here where boundary conditions are known at every time step, while only the initial velocity is known. The backward Euler method was used, in reference to Raissi et al. (2018). Doing so, u^t can be expressed as

$$u^t = u^{t-1} + \frac{\Delta t}{\rho} (-\nabla p + \eta \nabla^2 u + f_{ext}) \quad (10)$$

Rearranging the equation yields,

$$u^t + \frac{\Delta t}{\rho} (\nabla p - \eta \nabla^2 u) = u^{t-1} + \frac{\Delta t}{\rho} f_{ext} \quad (11)$$

Here, the *LHS* of the equation is defined as $\mathcal{G}$ where

$$\mathcal{G} \equiv u^t + \frac{\Delta t}{\rho} (\nabla p - \eta \nabla^2 u) \quad (12)$$

substituting some terms with f , where f is defined as following,

$$f = \nabla p - \eta \nabla^2 u \quad (13)$$

we get the value of $\mathcal{G}$ as shown below.

$$\mathcal{G} \equiv u^t + \frac{\Delta t}{\rho} f$$

$$\mathcal{G} \equiv u^{t-1} + \frac{\Delta t}{\rho} f_{ext}$$

Equations (2.1) and (2.1) are two expressions for the same quantity: $\mathcal{G}$ can be evaluated from data at time $t - 1$ (the external force and the previous velocity), while it also appears, through f , as a linear combination of the unknown u^t and p^t . This dual role is what allows $\mathcal{G}$ to impose the physical constraint linking u^t to u^{t-1} within the Gaussian process framework developed next.

2.2 Physics-Informed Kernel Construction

Molina et al. (2023) developed the Stokesian process, in which the linearity of the steady Stokes equations is used to construct a Gaussian process kernel from the governing equations directly, together with the incompressibility constraint

$$\mathbf{s} = \nabla \cdot \mathbf{u} \quad (14)$$

For a channel flow problem, where the driving force is a single-axis pressure gradient, the velocity at the walls and the pressure at the inlet and outlet are known. Given training data and unknown test locations $\mathbf{X}_\star = [\mathbf{x}_{1,\star}, \mathbf{x}_{2,\star}, \dots, \mathbf{x}_{N,\star}]$, with unknown velocity and pressure denoted $\mathbf{u}^\star$ and $\mathbf{p}^\star$, the joint Gaussian process over s and $\mathcal{G}$ is

$$\begin{bmatrix} u^* \\ p^* \\ u \\ p \\ \mathcal{G} \\ s \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} K^{**} & Q \\ Q^T & K \end{bmatrix} \right) \quad (15)$$

$$K = \begin{bmatrix} K_{uu} & K_{up} & K_{uG} & K_{us} \\ & K_{pp} & K_{pG} & K_{ps} \\ & & K_{GG} & K_{Gs} \\ & & & K_{ss} \end{bmatrix} \quad (16)$$

In order to investigate inference of time propagation in velocity, we defined the test value to be value of the variable at time t , and the training value to be of time $t-1$. The appropriate prior can be re-written as following.

$$\begin{bmatrix} u^t \\ p^t \\ u^{t-1} \\ p^{t-1} \\ \mathcal{G}^t \\ s \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} K^{t,t} & Q \\ Q^T & K^{t-1,t-1} \end{bmatrix} \right) \quad (17)$$

$$K^{t-1,t-1} = \begin{bmatrix} K_{u^{t-1},u^{t-1}} & K_{u^{t-1},p^{t-1}} & K_{u^{t-1},G^t} & K_{u^{t-1},s} \\ & K_{p^{t-1},p^{t-1}} & K_{p^{t-1},G^t} & K_{p^{t-1},s} \\ & & K_{G^t,G^t} & K_{G^t,s} \\ & & & K_{s,s} \end{bmatrix} \quad (18)$$

$$Q = \begin{bmatrix} K_{u^t,u^{t-1}} & K_{u^t,p^{t-1}} & K_{u^t,G^t} & K_{u^t,s} \\ K_{p^t,u^{t-1}} & K_{p^t,p^{t-1}} & K_{p^t,G^t} & K_{p^t,s} \end{bmatrix} \quad (19)$$

$$K^{t,t} = \begin{bmatrix} K_{u^t,u^t} & K_{u^t,p^t} \\ K_{p^t,p^t} \end{bmatrix} \quad (20)$$

At this point, a separate kernel function would be required to specify each pairwise correlation among the variables appearing in Equation (18). However, since all of these variables are linearly related through the Stokes and continuity equations, every correlation function can instead be expressed

in terms of only $d(d+1)/2$ velocity–velocity correlations $k_{u^\alpha u^\beta}$, d velocity–pressure correlations $k_{u^\alpha p}$, and a single pressure–pressure correlation function k_{pp} , where d is the spatial dimension (Molina et al., 2023). For the two-dimensional flows considered here ($d = 2$), this amounts to three velocity–velocity kernels ($k_{u^x u^x}, k_{u^x u^y}, k_{u^y u^y}$), two velocity–pressure kernels ($k_{u^x p}, k_{u^y p}$), and the single pressure–pressure kernel k_{pp} – six base correlation functions in total. Because the covariance kernels depend only on spatial position and not explicitly on time, the same six base kernels also determine the cross-time correlations (e.g. between u^t and u^{t-1}) needed for the time-marching scheme of Section 2.1; only the assignment of a given set of values to the training or test role changes from one time step to the next. All remaining correlations, involving f^α , s , and $\mathcal{G}^\alpha$, are derived from these six base kernels by applying the linear differential operators of the governing equations, as follows, for $\rho = 1$.

$$k_{u^\alpha f^\beta} = \mathcal{D}'^\beta k_{u^\alpha p} - \eta \mathcal{L}' k_{u^\alpha u^\beta} \quad (21)$$

$$k_{u^\alpha s} = \mathcal{D}'^\beta k_{u^\alpha u^\beta} \quad (22)$$

$$k_{p f^\alpha} = \mathcal{D}'^\alpha k_{pp} - \eta \mathcal{L}' k_{p u^\alpha} \quad (23)$$

$$k_{ps} = \mathcal{D}'^\alpha k_{p u^\alpha} \quad (24)$$

$$k_{f^\alpha u^\beta} = \mathcal{D}^\alpha k_{p u^\beta} - \eta \mathcal{L} k_{u^\alpha u^\beta} \quad (25)$$

$$k_{f^\alpha f^\beta} = \mathcal{D}^\alpha \mathcal{D}'^\beta k_{pp} - \mathcal{D}^\alpha \eta \mathcal{L}' k_{p u^\beta} - \eta \mathcal{L} \mathcal{D}'^\beta k_{u^\alpha p} + \eta^2 \mathcal{L} \mathcal{L}' k_{u^\alpha u^\beta} \quad (26)$$

$$k_{f^\alpha s} = \mathcal{D}^\alpha \mathcal{D}'^\beta k_{p u^\beta} - \eta \mathcal{L} \mathcal{D}'^\beta k_{u^\alpha u^\beta} \quad (27)$$

$$k_{ss} = \mathcal{D}^\alpha \mathcal{D}'^\beta k_{u^\alpha u^\beta} \quad (28)$$

$$k_{u^\alpha \mathcal{G}^\beta} = k_{u^\alpha u^\beta} + \Delta t k_{u^\alpha f^\beta} \quad (29)$$

$$k_{p \mathcal{G}^\beta} = k_{p u^\beta} + \Delta t k_{p f^\beta} \quad (30)$$

$$k_{\mathcal{G}^\alpha \mathcal{G}^\beta} = k_{u^\alpha u^\beta} + \Delta t k_{u^\alpha f^\beta} + \Delta t k_{f^\alpha u^\beta} + (\Delta t)^2 k_{f^\alpha f^\beta} \quad (31)$$

$$k_{s \mathcal{G}^\beta} = k_{s u^\beta} + \Delta t k_{s f^\beta} \quad (32)$$

$$(33)$$

The general posterior is then

$$\begin{bmatrix} u^t \\ p^t \end{bmatrix} \Bigg| \begin{bmatrix} u^{t-1} \\ p^{t-1} \\ \mathcal{G}^t \\ s \end{bmatrix} \sim \mathcal{N} \left(\mathbf{Q} \mathbf{K}^{-1} \begin{bmatrix} u^{t-1} \\ p^{t-1} \\ \mathcal{G}^t \\ s \end{bmatrix}, \mathbf{K}_{t,t} - \mathbf{Q} \mathbf{K}^{-1} \mathbf{Q}^T \right) \quad (34)$$

2.3 Application to Periodic Channel Flow

Molina et al. (2023) have also shown that with periodic boundary conditions for the velocity u , together with a constant pressure difference Δp between the inlet ($x = 0$) and outlet ($x = L$)

$$u^\alpha(x + L, y) = u^\alpha(x, y) \quad (35)$$

$$p(x + L, y) = p(x, y) + \Delta p. \quad (36)$$

These can be expressed as $(S_L^1 u^\alpha)(0, y) = 0$ and $(S_L^1 p)(0, y) = -\Delta p$. For the periodic channel-flow geometry considered in this work, the generic training set of Equations (18)–(19) is specialized as follows: the full-field u^{t-1} and p^{t-1} training data are replaced by the time-invariant no-slip wall velocity $\mathbf{u}_b$ and the periodicity constraints $S_L^1 \mathbf{u}$, $S_L^1 \mathbf{p}$, while $\mathcal{G}^t$ and s are retained. Information about u^{t-1} enters the inference only through $\mathcal{G}^t$ (Equation (2.1)), not as a separate training block. $\mathbf{Q}$ and $\mathbf{K}^{t-1, t-1}$ are redefined analogously to Equations (18) and (19), with u^{t-1}, p^{t-1} replaced by u_b , $S_L^1 \mathbf{u}$, and $S_L^1 \mathbf{p}$. The prior becomes

$$\begin{bmatrix} u^t \\ u_b \\ S_L^1 \mathbf{u} \\ S_L^1 \mathbf{p} \\ \mathcal{G}^t \\ s \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} \mathbf{K}^{t,t} & \mathbf{Q} \\ \mathbf{Q}^T & \mathbf{K}^{t-1, t-1} \end{bmatrix} \right) \quad (37)$$

The posterior changes form to the following form.

$$\begin{matrix} u^t \\ p^t \end{matrix} \Bigg| \begin{bmatrix} \mathbf{u}_b \\ S_L^1 \mathbf{u} \\ S_L^1 \mathbf{p} \\ \mathcal{G}^t \\ s \end{bmatrix} \sim \mathcal{N} \left(\mathbf{Q} \mathbf{K}^{-1} \begin{bmatrix} \mathbf{u}_b \\ S_L^1 \mathbf{u} \\ S_L^1 \mathbf{p} \\ \mathcal{G}^t \\ s \end{bmatrix}, \mathbf{K}^{t,t} - \mathbf{Q} \mathbf{K}^{-1} \mathbf{Q}^T \right) \quad (38)$$

2.4 Uncertainty propagation

Following Raissi et al. (2018), since $\mathcal{G}^t$ is built from the previous step's posterior mean $\boldsymbol{\mu}^{t-1}$ rather than an exactly known value, its uncertainty

$\mathbf{u}^{t-1} \sim \mathcal{N}(\boldsymbol{\mu}^{t-1}, \boldsymbol{\Sigma}^{t-1,t-1})$ must be marginalized out to obtain a consistent predictive covariance. Doing so gives

$$\begin{bmatrix} \mathbf{u}^t \\ \mathbf{p}^t \end{bmatrix} \mid \mathbf{u}_b, S_L^1 \mathbf{u}, S_L^1 \mathbf{p}, \mathbf{s} \sim \mathcal{N}(\boldsymbol{\mu}^t, \boldsymbol{\Sigma}^{t,t}) \quad (39)$$

where

$$\boldsymbol{\mu}^t = \mathbf{Q} \mathbf{K}^{-1} \begin{bmatrix} \mathbf{u}_b \\ S_L^1 \mathbf{u} \\ S_L^1 \mathbf{p} \\ \boldsymbol{\mu}^{t-1} + \frac{\Delta t}{\rho} \mathbf{f}_{ext} \\ \mathbf{s} \end{bmatrix} \quad (40)$$

$$\boldsymbol{\Sigma}^{t,t} = \mathbf{K}^{t,t} - \mathbf{Q} \mathbf{K}^{-1} \mathbf{Q}^T + \mathbf{Q} \mathbf{K}^{-1} \mathbf{M} \mathbf{K}^{-1} \mathbf{Q}^T \quad (41)$$

and $\mathbf{M}$ is zero everywhere except in the block corresponding to $\boldsymbol{\mathcal{G}}^t$, where it equals $\boldsymbol{\Sigma}^{t-1,t-1}$, the previous step's posterior covariance evaluated at the artificial training locations. This is the propagated-uncertainty analogue of equation (13) in Raissi et al. (2018).

2.5 Hyperparameters Optimization

Following Molina et al. (2023), the kernel hyperparameters $\boldsymbol{\theta}$ are found via a maximum a posteriori (MAP) estimate rather than full Bayesian marginalization,

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} [\log p(\boldsymbol{\theta} \mid \mathbf{X}, f(\mathbf{X}), I)], \quad (42)$$

where $p(\boldsymbol{\theta} \mid \mathbf{X}, f(\mathbf{X}), I) \propto p(f(\mathbf{X}) \mid \mathbf{X}, \boldsymbol{\theta}, I) p(\boldsymbol{\theta} \mid I)$, following the same construction as Molina et al. (2023).

following Molina et al. (2023), we take $p(\boldsymbol{\theta} \mid I)$ to be an independent Jeffreys prior for each hyperparameter,

$$\log p(\boldsymbol{\theta} \mid I) = \sum_{i=1}^{N_{\theta}} \log p(\theta_i \mid I) = - \sum_{i=1}^{N_{\theta}} \log \theta_i. \quad (43)$$

The initial hyperparameter values $(\gamma_1, \gamma_2, \log l) = (1.2, 0, -1.2)$ are taken from Molina et al. (2023), where γ_1 is the initial variance scale for the u - u and p - p correlations, γ_2 the initial variance scale for u - p correlations, and l the initial length scale, shared across all correlations at initialization. Unlike Molina et al. (2023), where $\gamma_2 = 0$ is only the starting point of a full optimization run over all hyperparameters, here the u - p correlation kernel $k_{u\alpha_p}$ is fixed at zero throughout optimization, i.e. u and p are modeled as uncorrelated fields.

Hyperparameters are optimized using a Nelder–Mead downhill simplex method over 500 iterations, matching Molina et al. (2023). Hyperparameters are re-fit from scratch at every time step, rather than warm-started from the previous step’s optimum. A jitter term is added to the training covariance diagonal for numerical stability, as in Molina et al. (2023),

$$\mathbf{K} \rightarrow \mathbf{K} + \epsilon^2 \mathbf{I}, \quad \epsilon = 10^{-3}. \quad (44)$$

2.6 Inference Algorithm

The full time-marching inference procedure is shown in Figure 1. During the first run, at time $t = 0$, $\mathbf{u}(t = 0)$ and $\mathbf{p}(t = 0)$ are both zero. The training set consists of 340×2 points for $\mathcal{G}$, 340 points for s , 60×2 points for the no-slip wall velocity $\mathbf{u}_b$, 15×2 points for $S_L^1 \mathbf{u}$, and 15 points for $S_L^1 \mathbf{p}$, matching Molina et al. (2023). At each step, hyperparameters are optimized as described in Section 2.5. 500 test points are predicted, of which 340 are randomly sampled from the predicted output to serve as the artificial $\mathcal{G}$ training data for the next step. In every subsequent loop, the number

of training points remains the same as in the first round, but $\mathcal{G}^t$ is recomputed by substituting the posterior mean of u^{t-1} from the previous loop into Equation (2.1), with its uncertainty propagated as described in Section 2.4.

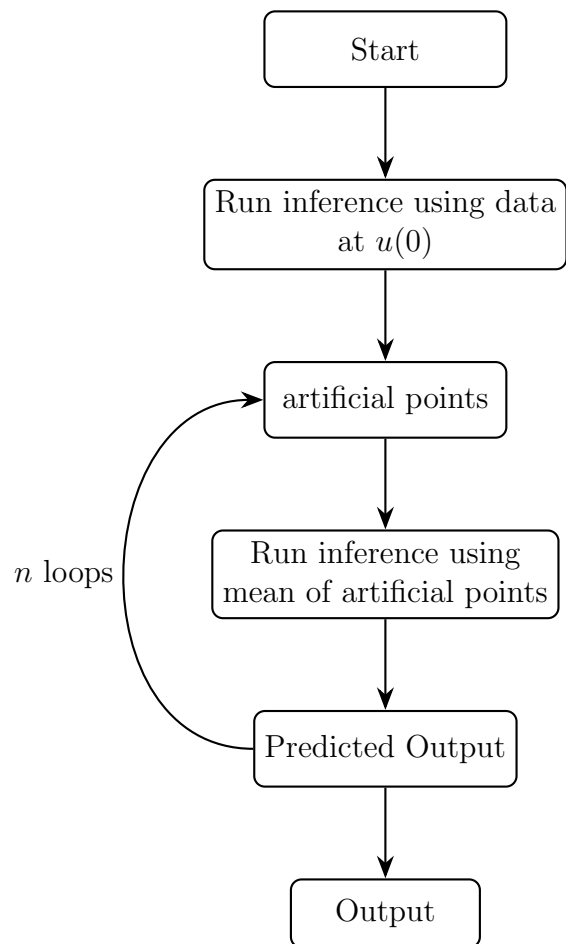


Figure 1: Flowchart of the predictive simulation loop.

Chapter 3

Results

This chapter presents the inferred velocity field for pressure-driven, unsteady Stokes flow in two two-dimensional geometries, using the time-marching physics-informed Gaussian process developed in Chapter 2. Section 3.1 considers flow between two parallel plates, where a closed-form analytical solution is available for validation. Section 3.2 considers flow through a sinusoidal channel, following the geometry of Molina et al. (2023), where a finite-element (FEM) solution is instead used as the reference.

3.1 Pressure-driven flow between parallel plates

For an impulsively started plane Poiseuille flow between two stationary plates, initially at rest, with a pressure gradient dp/dx suddenly applied at $t = 0$, the unsteady Stokes equations (2) admit the closed-form series solution of Muñoz et al. (2012) in section 4.3. Writing the velocity as the sum of the steady plane Poiseuille profile and a decaying transient part, $u_x(y, t) = u_0(y) + u_e(y, t)$, and specializing their result to two stationary plates ($U = 0$) over the domain $y \in [-0.5, 0.5]$ used in this work (via the shift $y' = y + 0.5 \in [0, 1]$),

$$u_x(y, t) = -\frac{y'(1-y')}{2\eta} \frac{dp}{dx} - \frac{2}{\eta} \frac{dp}{dx} \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{4}{(n\pi)^3} e^{-n^2\pi^2\nu t} \sin(n\pi y'), \quad (45)$$

$$y' = y + 0.5, \quad \nu = \eta/\rho, \quad (46)$$

which converges to the steady plane Poiseuille profile as $t \rightarrow \infty$. Here,

the value of $dp/dx = -12.0$. This analytical solution provides a direct, independent check of the PIGP solution.

The time step was $\Delta t = 0.05$, with ten time-marching loops from $t = 0$, using 270 artificial training points per step (Section 2.6). Figure 2 shows the predicted u_x field at five time snapshots, compared against Equation (46).

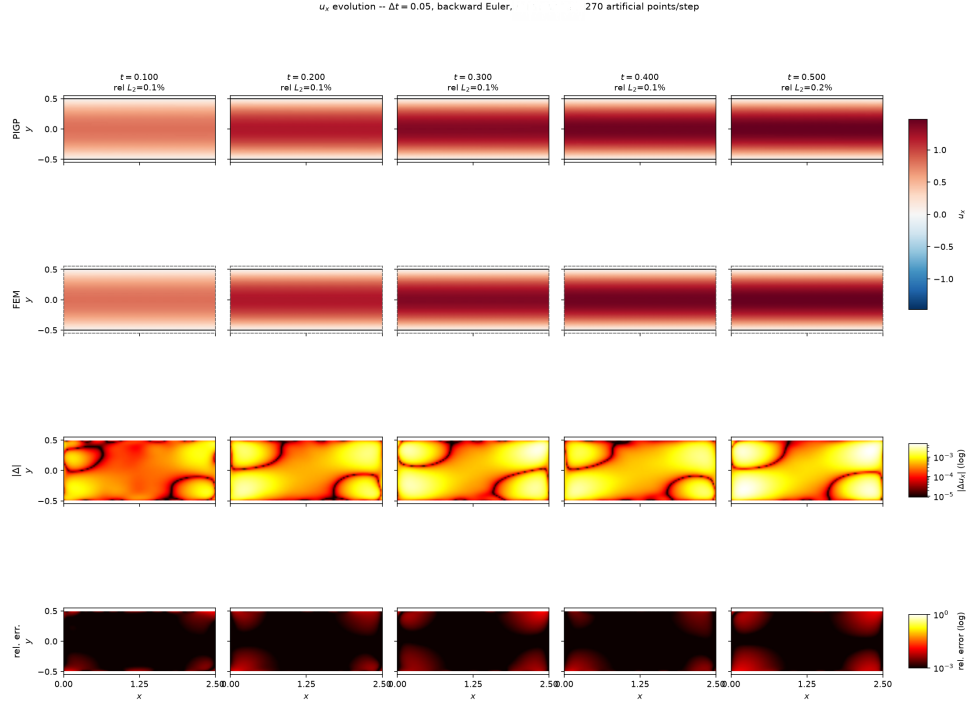


Figure 2: Evolution of u_x for pressure-driven flow between parallel plates, $\Delta t = 0.05$, 10 loops, 270 artificial points per step, compared against the analytical solution of Equation (46). Top row: PIGP posterior mean. Middle row: analytical solution. Bottom two rows: absolute and relative error.

The relative L_2 error remains at 0.1–0.2% across all five snapshots ($t = 0.1$ to $t = 0.5$), with no clear growth trend over the ten loops, indicating that error does not compound significantly over this time horizon at this step size. The largest local errors, visible in the third and fourth rows of Figure 2, are concentrated near the channel walls ($y \rightarrow \pm 0.5$) rather than at the channel centerline. This is consistent with the no-slip boundary condition

producing the sharpest velocity gradients in u_x near the walls, where the squared-exponential kernel’s finite length scale is least able to resolve the local curvature of the profile relative to the smoother, near-parabolic region at the channel core.

3.2 Pressure driven flow through a sinusoidal channel

Following Molina et al. (2023), velocities for pressure-driven flow through a sinusoidal channel of width

$$w(x) = \langle w \rangle + 2a \sin(2\pi x/L) \quad (47)$$

were predicted and compared against a finite-element (FEM) reference solution, using both the absolute error $|\Delta u| = |u_{\text{PIGP}} - u_{\text{FEM}}|$ and the relative error $|\Delta u|/|u_{\text{FEM}}|$ to assess accuracy. The time step was $\Delta t = 0.05$, with ten time-marching loops from $t = 0$, using 340 artificial training points per step (Section 2.6).

Figures 3 and 4 show the predicted u_x and u_y fields at five time snapshots, compared against FEM.

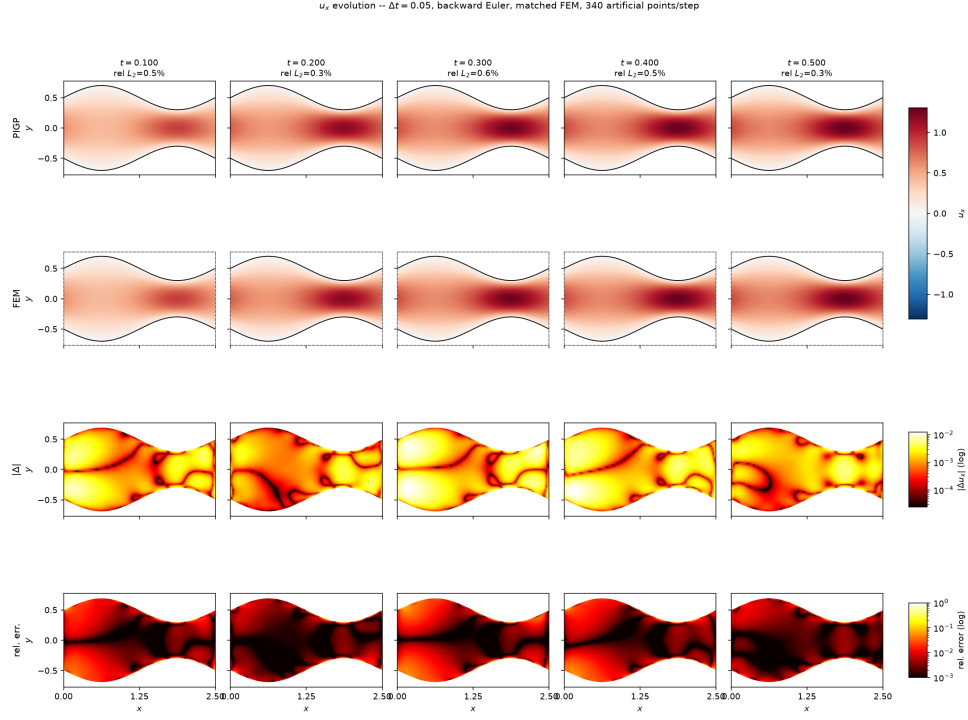


Figure 3: Evolution of u_x for pressure-driven flow through a sinusoidal channel, $\Delta t = 0.05$, 10 loops, 340 artificial points per step. Top row: PIGP posterior mean. Middle row: FEM reference. Bottom two rows: absolute and relative error.

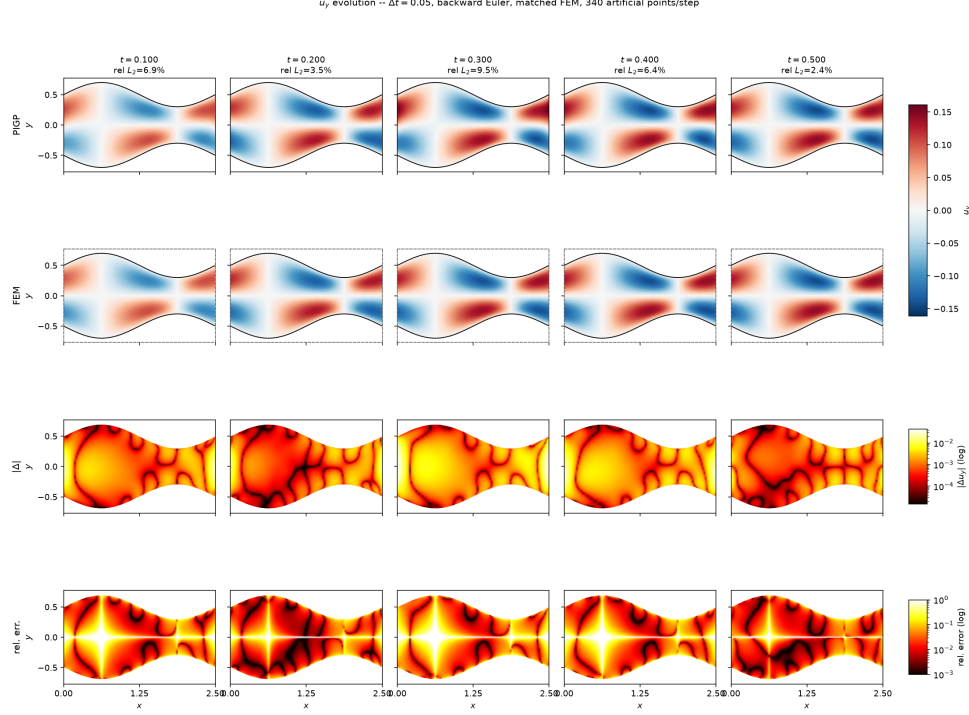


Figure 4: Evolution of u_y for pressure-driven flow through a sinusoidal channel, $\Delta t = 0.05$, 10 loops, 340 artificial points per step. Top row: PIGP posterior mean. Middle row: FEM reference. Bottom two rows: absolute and relative error.

The relative L_2 error in u_x is non-monotonic over the five snapshots, peaking at 0.6% at $t = 0.3$, before falling to 0.3% by $t = 0.5$. This is consistent with the flow’s transient adjustment: early times remain close to the exactly-known zero-velocity initial condition, so error is naturally small; as the flow develops and accelerates through the narrowest point of the channel, spatial velocity gradients become sharper and harder for the finite-length-scale kernel to resolve, peaking around $t = 0.3$; as the flow approaches its quasi-steady state at later times, gradients stabilize and the error recedes.

Errors are consistently larger for u_y than for u_x across all five snapshots. This is expected on physical grounds rather than being a shortcoming of the PIGP inference itself: in a mildly-perturbed sinusoidal channel, u_x is the dominant, order-one velocity component along the channel axis, while u_y cap-

tures only the smaller cross-stream motion induced by the wall’s curvature. Since relative error normalizes the absolute error by the local magnitude of the reference field, the same absolute prediction error corresponds to a substantially larger relative error in u_y , simply because $|u_y| \ll |u_x|$ throughout most of the domain. This is also consistent with the errors in this section being larger overall than those reported for the parallel-plate case in Section 3.1 (0.1–0.2%), reflecting the added geometric complexity of the sinusoidal wall.

3.2.1 Effect of Time Step Size

To test the effect of the time discretization, the sinusoidal-channel case of Section 3.2 was repeated with a smaller time step, $\Delta t = 0.01$, run for 25 loops ($t \leq 0.25$), keeping 340 artificial training points per step.

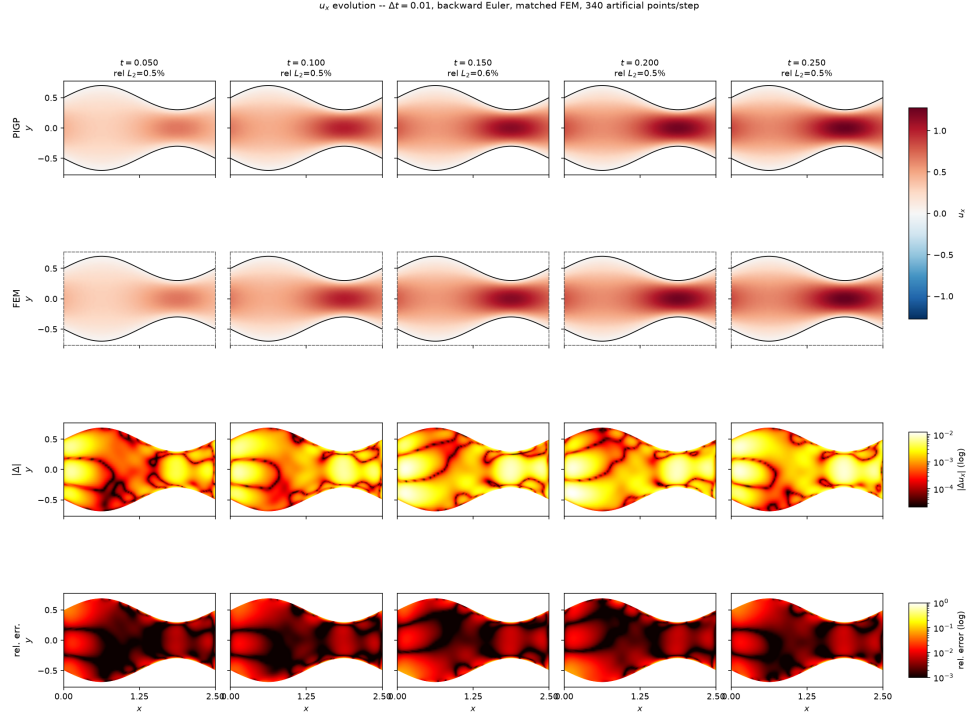


Figure 5: Evolution of u_x for pressure-driven flow through a sinusoidal channel, $\Delta t = 0.01$, 25 loops, 340 artificial points per step. Top row: PIGP posterior mean. Middle row: FEM reference. Bottom two rows: absolute and relative error.

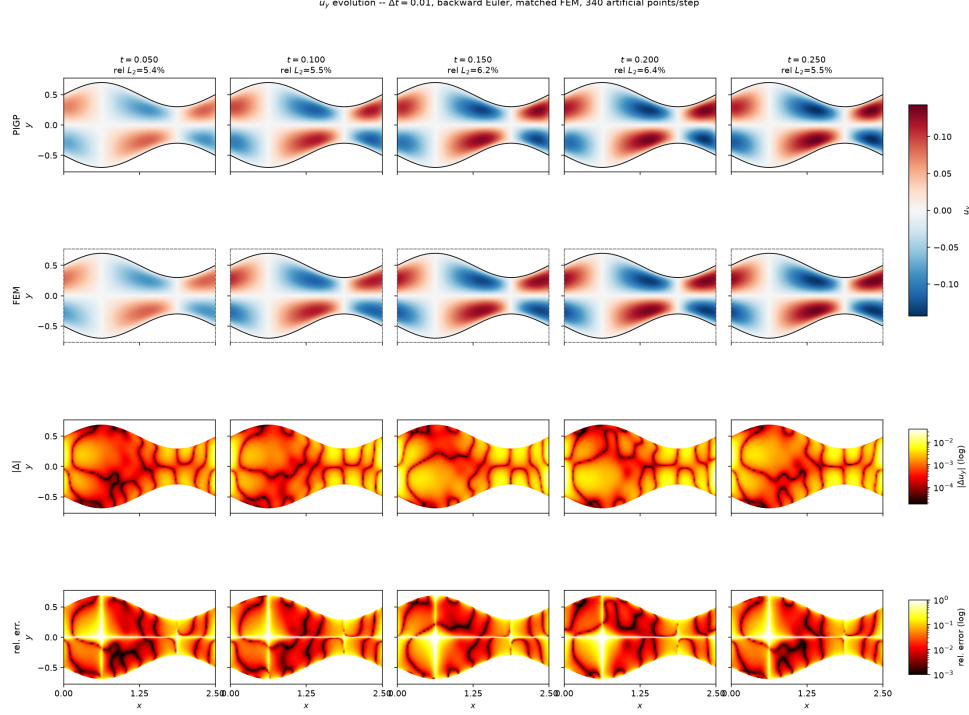


Figure 6: Evolution of u_y for pressure-driven flow through a sinusoidal channel, $\Delta t = 0.01$, 25 loops, 340 artificial points per step. Top row: PIGP posterior mean. Middle row: FEM reference. Bottom two rows: absolute and relative error.

At the one snapshot common to both runs, $t = 0.1$, the smaller time step ($\Delta t = 0.01$) gives a smaller u_y relative L_2 error than the larger time step ($\Delta t = 0.05$, Section 3.2), consistent with the expectation that a finer time discretization better approximates the continuous unsteady Stokes equations. However, this comparison cannot yet be extended to later times on equal footing: the $\Delta t = 0.05$, 10-loop run does not include a snapshot at $t = 0.25$, so no directly comparable data point exists there; whether the smaller-time-step advantage observed at $t = 0.1$ persists, narrows, or reverses at later times remains inconclusive from the runs available so far. Resolving this requires additional runs with matching snapshot times across step sizes, and extending the $\Delta t = 0.01$ run to a comparable total integration time (currently $t = 0.25$ versus $t = 0.5$ for the coarser run).

3.2.2 Effect of Artificial Training Data Size

To test the effect of the number of artificially generated training points per step, the sinusoidal-channel case was repeated with $N_{\text{art}} = 150, 220, 270$, and 340, holding $\Delta t = 0.05$ and 10 loops fixed. The relative L_2 errors are summarized in Table 1.

Table 1: Relative L_2 error (%) versus number of artificial training points per step, $\Delta t = 0.05$, 10 loops.

N_{art}	$t = 0.1$	0.2	0.3	0.4	0.5	mean
u_x						
150	4.5	3.6	3.2	5.3	3.1	3.94
220	0.6	0.4	0.9	0.3	0.4	0.52
270	0.6	0.4	0.4	0.3	0.3	0.40
340	0.5	0.3	0.6	0.5	0.3	0.44
u_y						
150	26.4	23.2	4.9	19.3	14.4	17.64
220	5.2	3.2	13.9	3.2	4.2	5.94
270	6.3	5.1	6.3	5.1	3.6	5.28
340	6.9	3.5	9.5	6.4	2.4	5.74

The initial expectation was that a larger number of training points would yield monotonically higher accuracy. The results support this only across the lower part of the range, and show clear saturation above $N_{\text{art}} \approx 220$.

At $N_{\text{art}} = 150$, the method performs poorly: the mean u_x error of 3.94% is roughly an order of magnitude larger than at any higher setting, and the mean u_y error of 17.6% is three times larger. Inspection of the predicted u_y fields themselves (Figure 8) shows this is not merely a loss of numerical precision: at $t = 0.4$ the predicted vortex structure is visibly distorted relative to the FEM reference, indicating that 150 points is below the threshold required to constrain the governing equations over this geometry at all.

Above this threshold the returns diminish sharply. Increasing from 150 to 220 points reduces the mean u_x error from 3.94% to 0.52%, but the further increases to 270 and 340 points change it only to 0.40% and 0.44%. The same holds for u_y , where 220, 270 and 340 points give mean errors of 5.94%, 5.28% and 5.74% respectively. In both components the three larger settings differ by less than the snapshot-to-snapshot variation within any single run, so they should be regarded as equivalent in accuracy rather than ranked. Beyond

roughly 220 points, the quantity of artificial training data is no longer the accuracy-limiting factor for this problem.

A separate pattern is visible across the columns rather than the rows. In at least one component of each run: it is the maximum of its row for both components at 220 points (0.9% and 13.9%) and at 340 points (0.6% and 9.5%), and the joint maximum in u_y at 270 points (6.3%). Since this occurs independently of training set size, and in both velocity components, it is attributable to the state of the flow at that instant rather than to any one run: $t = 0.3$ falls within the transient during which the flow accelerates through the narrowest section of the channel, where the velocity gradients are sharpest and hardest for a finite-length-scale kernel to resolve. The 150-point run does not follow this pattern, but its errors are dominated by its overall failure to resolve the field and so are not informative here.

On this basis, $N_{\text{art}} = 270$ is a reasonable working choice for subsequent simulations. It matches the accuracy of the 340-point run in both components while using approximately 80% of the training points, and it is the setting least affected by the difficult $t = 0.3$ step. The practical benefit is a smaller covariance matrix to factorize at every time step.

These conclusions rest on a single geometry, a single time step, and a horizon of $t \leq 0.5$. Whether the saturation point shifts with channel geometry, with Δt , or over longer integration times, where accumulated time-marching error may place different demands on the training set, remains to be established, and simulations over a longer time frame are needed.

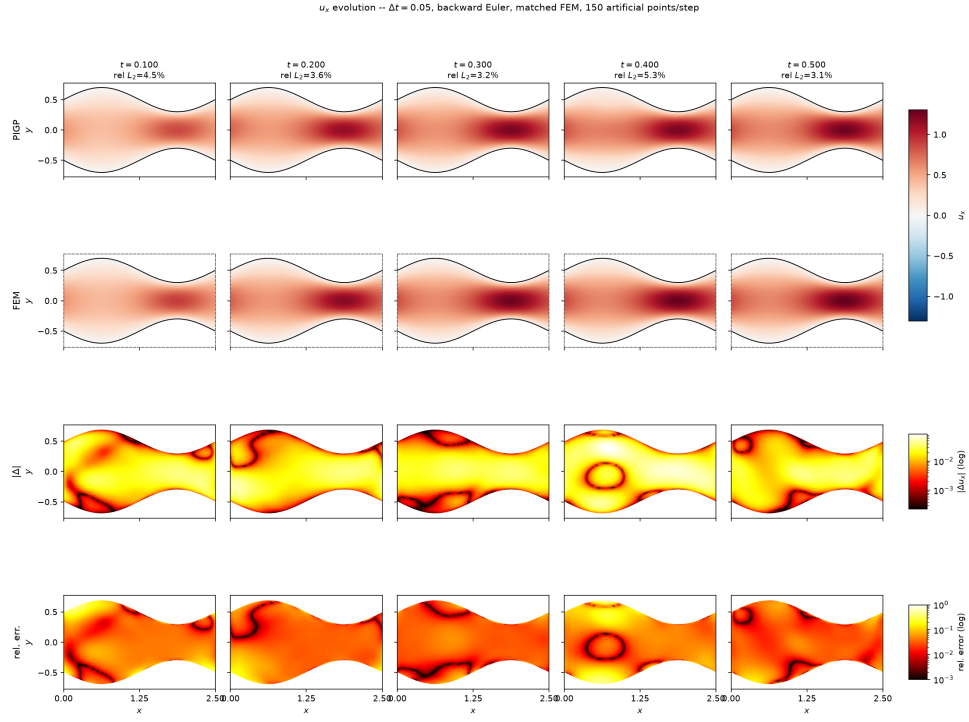


Figure 7: Evolution of u_x , 150 artificial points per step.

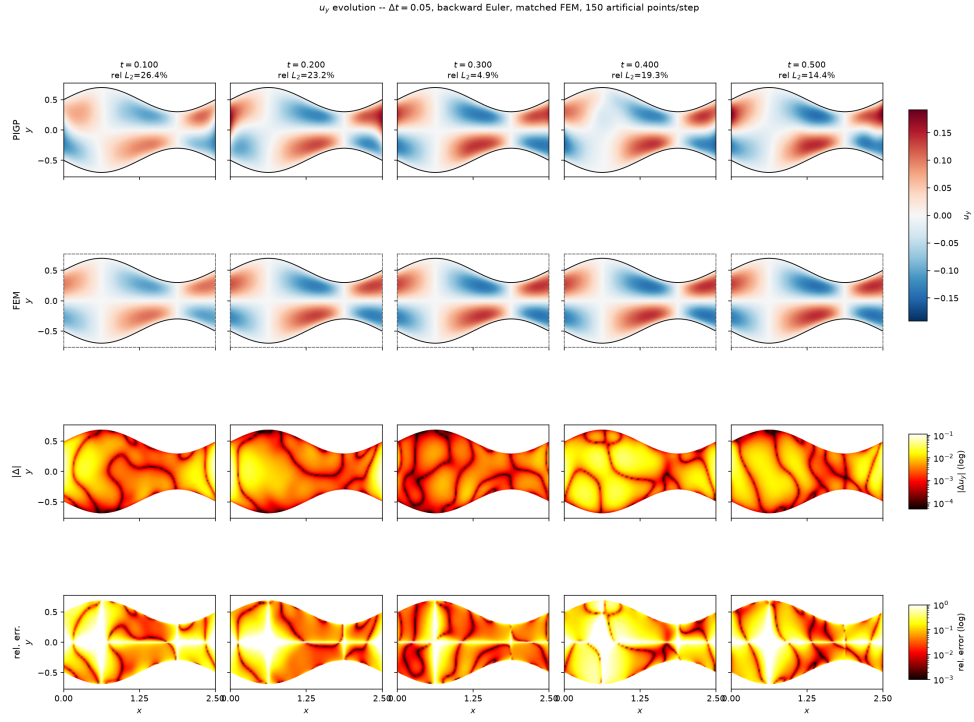


Figure 8: Evolution of u_y , 150 artificial points per step.

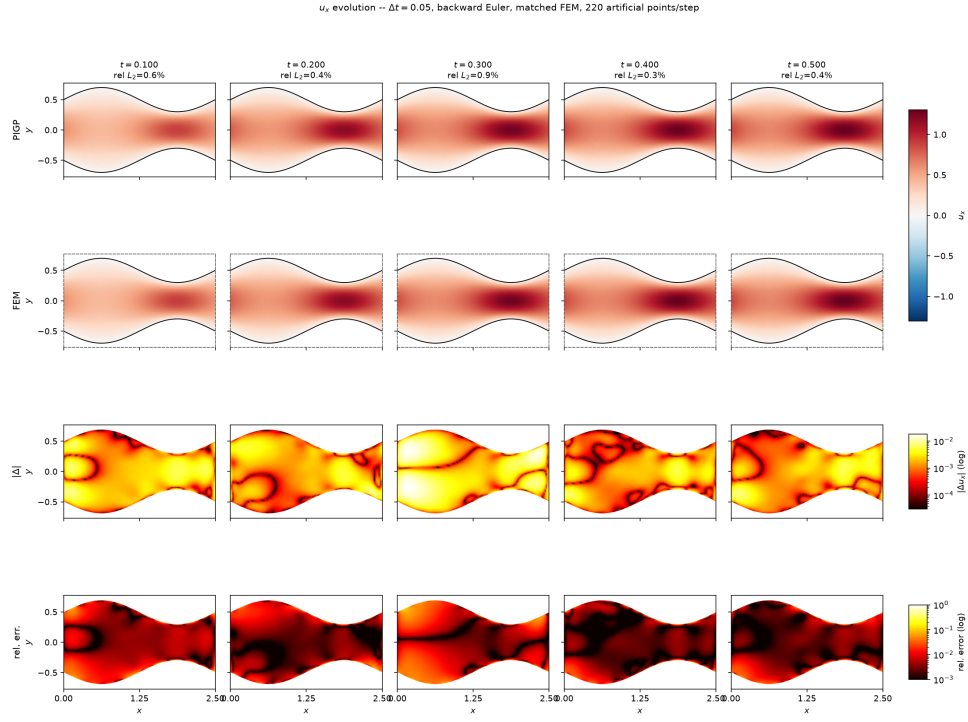


Figure 9: Evolution of u_x , 220 artificial points per step.

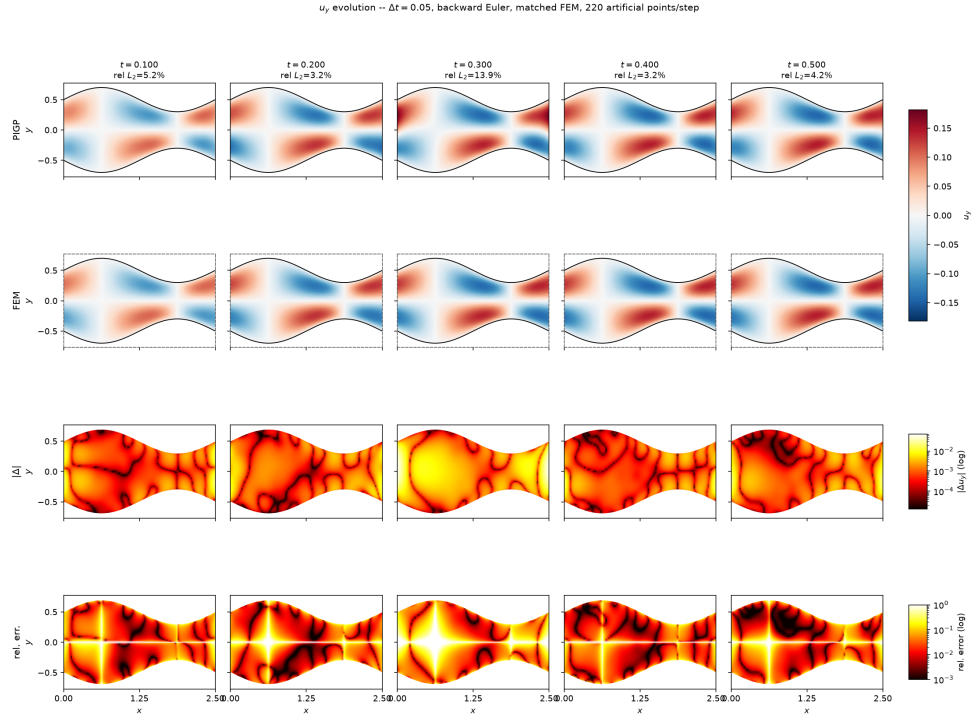


Figure 10: Evolution of u_y , 220 artificial points per step.

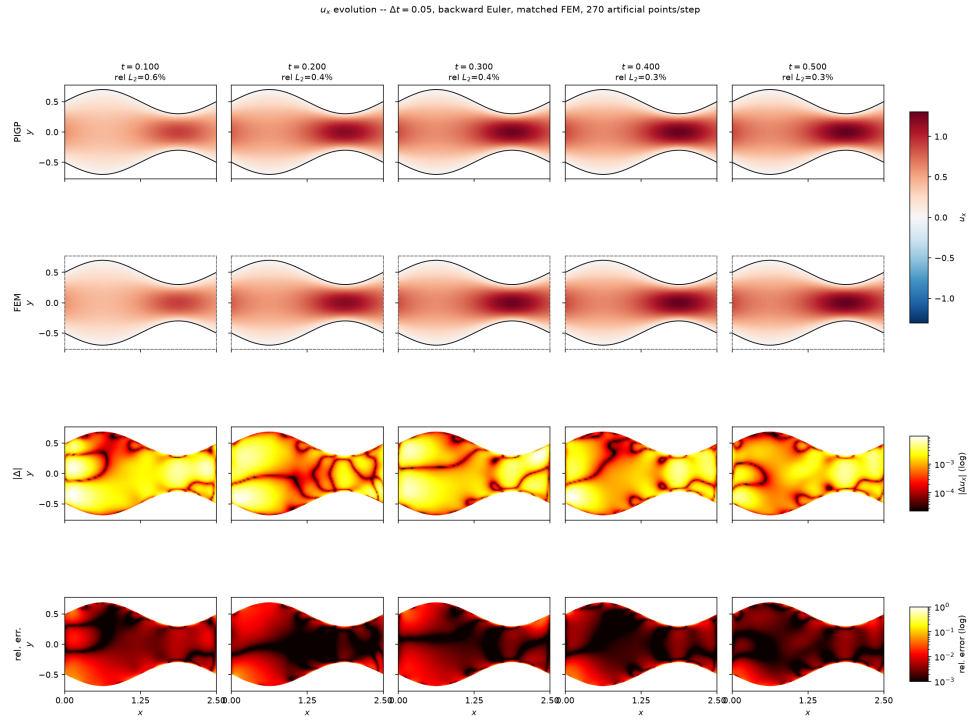


Figure 11: Evolution of u_x , 270 artificial points per step.

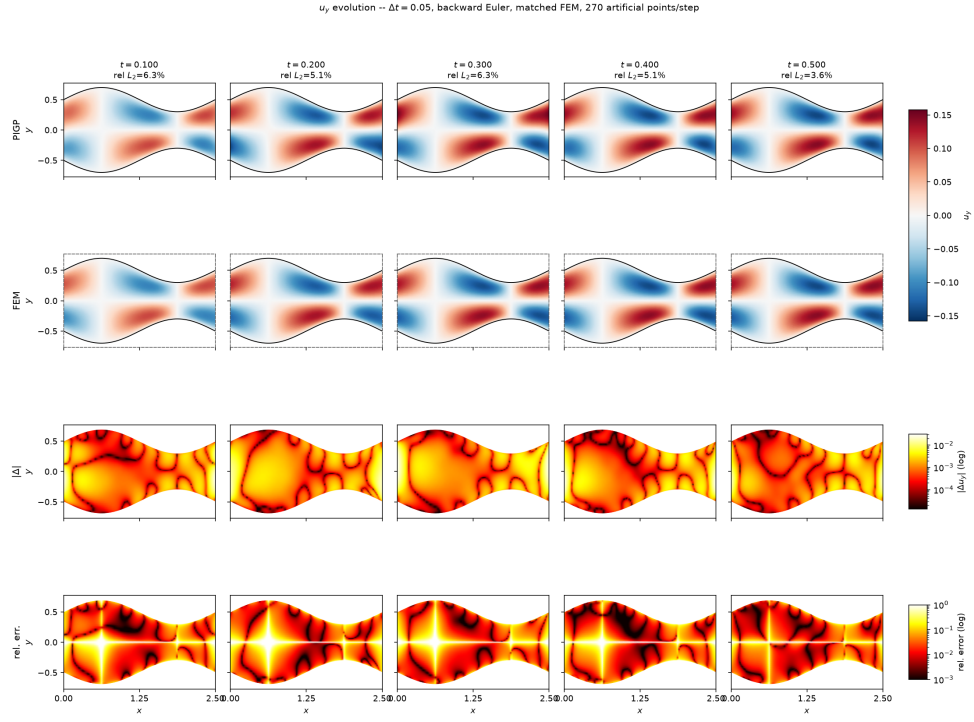


Figure 12: Evolution of u_y , 270 artificial points per step.

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